

KidneyGrid — demo run-book

Flower Collaborative Agent Hackathon · Stanford · Sept 29, 2026 · demos 5:30 pm · ~4 minutes

Links Web app: <http://127.0.0.1:8765> · Hub: flower.ai/apps/kdotmahesh/kidneygrid · GitHub: github.com/bouncypitch/kidneygrid · Federation: @kdotmahesh/kidneygrid · Submit (by 5:15): flowerlabs.typeform.com/to/rQuplUGG

1. Thirty minutes before (about 5:00)

- **Laptop:** plugged in, never sleep, on venue Wi-Fi, notifications off.
- **Nodes up:** `grep -l "SuperNode ID" .stack/grid-*.log | wc -l` should print **5** (Capitol Hospital runs on Nebius). If fewer: `scripts/supergrid_nodes.sh` and wait ~90 s.
- **Nebius:** `console.nebius.com` → Endpoints → *kidneygrid-capitol-hospital* shows **Running**.
- **Tabs, in order:** (1) deck in present mode · (2) web app, press **L** for Live · (3) `opener.mp4`.
- **Warm-up:** press **1** once in the web app so SuperGrid is warm. Browser zoom 100% at projector resolution.

2. The pitch

Time	Do	Say (short version)
0:00	Play opener video (40 s)	(let it play)
0:40	Slides 2–3: stakes, Maria	“95,000 Americans are waiting for a kidney. Maria has waited 7 years. Her husband can’t donate: wrong blood type.”
1:05	Slides 4–5: swap, silos	“Swaps work — Stanford’s Alvin Roth shared a Nobel for them. But hospitals can’t share patient data, so the matches are never found.”
1:25	Slide 6, then switch to web app	“KidneyGrid puts a Flower agent inside every hospital.”
1:35	Press 2 (Approve)	“Five hospitals and a courier on Flower’s SuperGrid — Capitol Hospital is running on Nebius. Alone: 0. First-come: 3, Maria left out. KidneyGrid: 5.” Point at the privacy ledger and a surgeon briefing : “Endeavor 1.0 writes each surgeon’s briefing, inside the hospital.”
2:35	Press 5 (OR cancels)	“Valley Health loses its operating room. The agents move the whole loop from Tuesday to Thursday, and a courier books a cold-chain van for every kidney without seeing a patient.”
3:15	Press 4 (Attack)	“Prompt injection is refused — privacy is enforced in code, not in a prompt.”
3:25	Press S (scale), back to deck	“On 300 simulated pairs: +41% transplants and 2.4× the hardest-to-match patients.”
3:40	Slides: Built on Flower, Path forward	“Native Flower: SuperNodes, Grid messaging, Flower Hub. Next: a real pilot. Zero matches alone. Five together. ”

3. If something goes wrong

- **Live run slow or stalls:** press **L** for Replay, press the same key again. Identical visuals, no network. Say: “Here’s the recorded run from 20 minutes ago.”
- **Wi-Fi dies:** Replay mode, plus the full film (`video/out/kidneygrid.mp4`).
- **A node dropped:** don’t fix it on stage — use Replay.

4. Likely judge questions

Question	Answer
Is this real patient data?	No — all synthetic. The architecture is real: each hospital's node reads only its own disk.
Doesn't HLA typing leave the hospital?	Only as anonymous donor tokens with no identity, as national registries already do. Antibody profiles and records never leave.
How does it scale?	Stanford fragmentation research estimates +30–63%; our simulation shows +41%.
Why Flower?	Data stays where it lives and agents collaborate across organizations — exactly what SuperNodes and the Grid are for.
Where is Endeavor used?	Each hospital's agent uses Endeavor 1.0 to write its surgeon's briefing from non-identifying facts.

5. Running KidneyGrid from Flower itself (for judges who ask)

- **Browser:** `flower.ai/app` → choose federation `@kdotmahesh/kidneygrid` → New chat → pick agent **KidneyGrid** → type “Approve the plan” (or “... an operating room may cancel”).
- **Terminal:** `uv run flwr chat → /federation @kdotmahesh/kidneygrid → @kdotmahesh/kidneygrid`
Approve the plan.
- The Hub page's **Run App** button fails for all AgentApps (Flower requires a chat prompt) — use chat instead. In a federation without hospital nodes, KidneyGrid runs a clearly labelled simulation.

6. After demos

- **Delete the Nebius endpoint** (Endpoints → the ... menu → Delete) — it bills about \$1.59/hour.
- Rotate the Flower API key that was shared in chat. Run `claude update`.